Scheme of valuation
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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-BCA (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Core Course (CC) - MG3CCRBCA202 - SOFTWARE ENGINEERING
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes Maximum Marks: 50

Part A

Very Short Answer Questions. Answer all questions

Each question carries 2 Marks

1. List any two key practices of Extreme Programming (XP). [K] [CO1]

Incremental planning, Small releases, Simple design, Test-first development, Refactoring, Pair programming, Continuous integration etc. . Two marks for writing any two practices

2. List some of the agile development tools. [K] [CO1]

Ontime developed by Axosoft - provides agile process management, Ideogramic UML - UML tool, Together tool set - supports technical activities within XP and other agile processes, Mentioning about collaborative and communication tools also can be considered, Jira, Trello, VersionOne, Asana, Rally ,Microsoft Azure DevOps etc.. Two marks for listing any two.

3. How has software shifted from product-based to service-based? [U] [CO1]

Shift from product purchase to online service model using SaaS and cloud computing. Users pay subscription, access anytime, get automatic updates, flexibility, and scalability.

4. Who is a Scrum master? [K][CO1]

A Scrum Master is the facilitator or the leader of the Scrum team who ensures that Scrum principles and practices are followed. They remove obstacles, guide the team, and promote collaboration for smooth project progress.

5. What is a risk exposure? [An] [CO2]

Risk exposure - is the overall consequence of a risk.

It is the potential loss or impact a project may face if a risk occurs. It is calculated as:

RE = PXC, Where P is the probability of occurrence of a risk and C is the cost to the project should the risk occur.

Risk Exposure = Probability × Impact.

Consider any one explanation (1 mark) and one formula (1 mark) .

6. What are the two stages of planning in agile development methods? [K] [CO2]

Release Planning, Iteration (Sprint) Planning, Daily Planning ,etc, (Any two ,each carries 1 mark)

7. Name two key characteristics of a good requirement. [An][CO2]

Clear and unambiguous, Complete, Consistent, Testable/verifiable, Feasible, Traceable, etc. (Any two, one mark each)

8. State the meaning of system testing. [U][CO3]

System testing is the process of testing the complete and integrated software system to ensure it meets the specified requirements and functions correctly as a whole.

9. What do you mean by the design concept "modularization" ? [U] [CO3]

Modularization is a design concept in which a software system is divided into separate, independent modules, each performing a specific function, to improve maintainability, re-usability, and understandability.

10. What are the two main aspects of software quality? [K] [CO3]

Functional quality**,** Non-functional quality, Product operation, product transition, Product revision Internal quality factors, external quality factors, Consider any one of the above three categories or any other meaningful answer. [2x10=20]

Part B

Short Answer Questions Answer any 3 questions

Each question carries 5 Marks

11. Describe the main phases of the Unified Process (UP) [K] [CO1]

Phases:- Inception **,** Elaboration Construction **,** Transition, Production

OR Communication, Planning, Modeling, Construction, Deployment,

Definition or listing the phases (2 Mark), Explanation -3Marks.

12. Explain agile software development. [U][CO1]

Definition/introduction, benefits (1Mark), Principles (2 Marks),

Examples/Models with features (2 Mark).

13. Compare the importance of each layer and argue which one is the most critical in ensuring software quality. [An][CO1]

Layered approach to software engineering Layers. (Tools, Method, Process, quality)(2 Marks), Importance of Each Layer (2 Marks), quality assurance provided by each Layer (1 Mark):

14. Explain the different steps involved in project scheduling process. [U] [CO2]

Identify activities - Identify activity dependencies - Estimate resources for activities -Create project charts

Listing the steps or processes (1 Marks) Explanation with diagram (3+1) Marks.

The explanation may include : - the process to Develop the Schedule, Create a project timeline using tools like Gantt charts or PERT/CPM charts to visualize task order and duration. Monitor and Control the Schedule, Track progress regularly, compare actual vs. planned performance, and make adjustments to stay on schedule.

15. Describe the importance of metrics for analysis models. [U] [CO3]

Introduction (1 Mark), Importance of Metrics for analysis / requirements model (4 Marks)

[5x3 = 15]

Part C

Essay Questions Answer any 1 question

Each question carries 15 Marks

16. Explain the role of requirements management throughout the software development life cycle. Discuss the importance of traceability and change control in managing requirements effectively. [An] [CO2]

Introduction (2 Marks), Role of Requirements Management in Each SDLC Phase (8 Marks), Importance of Traceability (3 Marks), Importance of Change Control (2 Marks)

17. Analyze common debugging strategies. How do systematic debugging methods help in reducing development time? [An] [CO3]

Introduction - the debugging process (3 Marks), Common Debugging Strategies -explanation and comparison of any two (6Marks), Systematic Debugging Methods and their Importance (4 Marks), effect of systematic debugging in reducing Development Time (2 Marks)

[15x1 = 15]

References:

1. Software Engineering A practitioner's Approach, 8th edition, Roger S Press man, Bruce R. Maxim. McGraw Hill Education, 2015.

2. Software Engineering, Ian Somerville, 9th edition, Pearson education